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IEEE Chandigarh Subsection International Conference : Submission (1062) has been created.

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Hello,

The following submission has been created.

Track Name: Artificial Intelligence and Data Science

Paper ID: 1062

Paper Title: Efficient Model Compression Techniques For Deploying Deep Neural Networks On Resource-Constrained Edge Devices

Abstract:
The rapid advancement of deep learning has enabled significant breakthroughs in areas such as computer vision, natural language processing, and speech recognition. However, deploying deep neural networks (DNNs) on resource-constrained edge devices remains a major challenge due to their high computational complexity, memory requirements, and energy consumption. Edge devices, including smartphones, IoT devices, and embedded systems, often lack the hardware capabilities required to efficiently run large-scale deep learning models.
This paper presents a comprehensive study of efficient model compression techniques aimed at enabling the deployment of deep neural networks on resource-constrained edge devices. We explore key compression methods, including pruning, quantization, knowledge distillation and low-rank factorization. Furthermore, we propose an integrated compression pipeline that combines multiple techniques to achieve optimal trade-offs between model size, accuracy, latency, and energy efficiency.
Extensive analysis demonstrates that hybrid compression approaches can significantly reduce model size while maintaining acceptable performance levels. The findings highlight the importance of balancing efficiency and accuracy, making compressed models suitable for real-time edge applications.

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Submission Files:
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Submission Questions Response:
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